

4. Footing segment

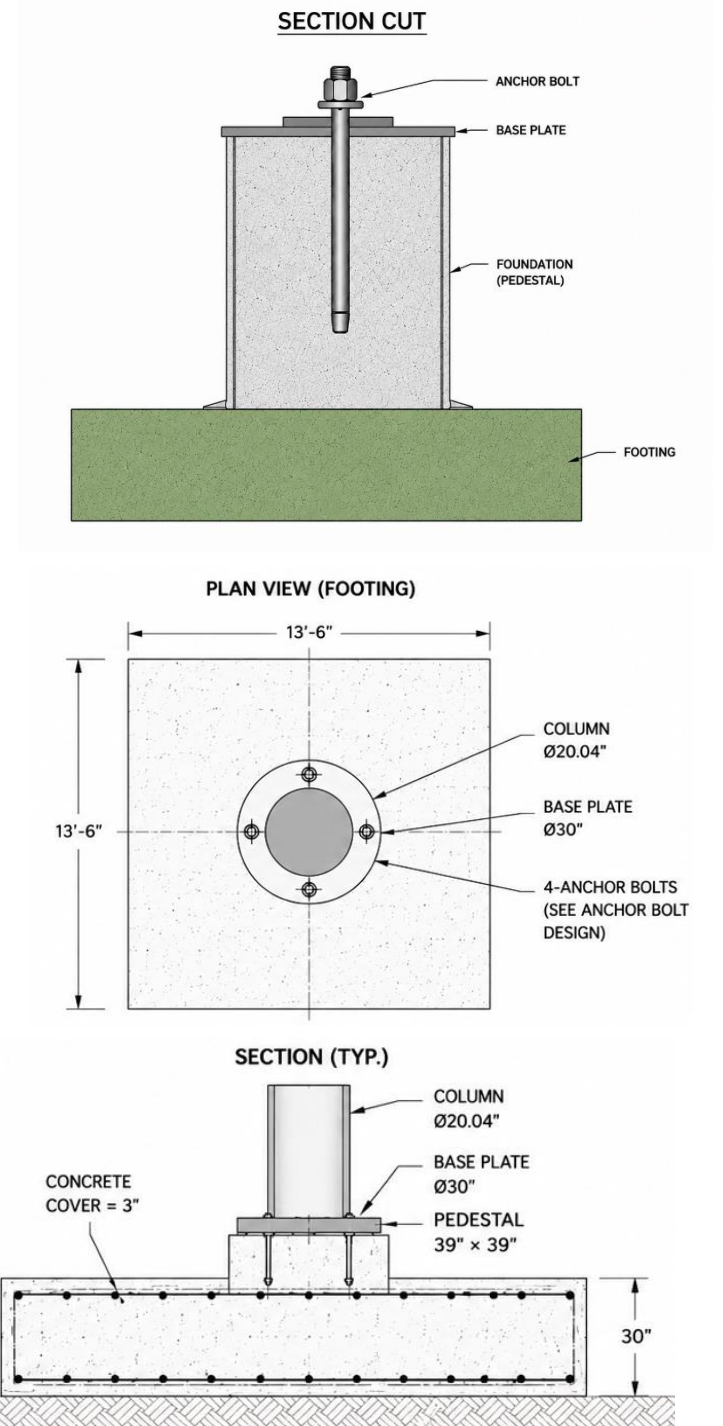
No	Design Parameter	Symbol	Value	Source
1	Total factored axial load	P_u	1609.756 kip	Main Tank Design
2	Factored overturning moment	M_u	5673.437 kip-ft	Main Tank Design
3	Factored shear	V_u	52.30 kip	Main Tank Design
4	Number of legs	N_{leg}	4	Tank Geometry
5	Pedestal size	$B \times L$	39×39 in	Pedestal Design
6	Pedestal height	H_p	7'-5"	Pedestal Design
7	Concrete strength	f_c	4 ksi	User Input
8	Reinforcing steel	f_y	60 ksi	User Input
9	Allowable soil pressure (Assumed)	q_{allow}	3.0 ksf	User Input

Step 1: Footing Load

Gravity load per leg

$$P_{leg} = \frac{P_u}{N_{leg}}$$

$$P_{leg} = \frac{1609.756}{4} = 402.44 \text{ kip}$$



The governing compression-side overturning reaction is:

$$T_u = 103.65 \text{ kip}$$

Therefore:

$$\begin{aligned} P_c &= P_{leg} + T_u \\ P_c &= 402.44 + 103.65 = 506.09 \text{ kip} \end{aligned}$$

For footing design, including the pedestal self-weight allowance adopted in the design:

$$P_u = 520 \text{ kip}$$

This value is used consistently throughout the footing strength calculations.

Step 2: Preliminary Footing Size

Required footing area

$$\begin{aligned} A &= \frac{P}{q_{allow}} \\ A &= \frac{520}{3} = 173.3 \text{ ft}^2 \end{aligned}$$

Adopt

$$B = L = 13.5 \times 13.5 \text{ ft}$$

Area

$$A_f = 13.5(13.5) = 182.25 \text{ ft}^2$$

The adopted footing area is adequate based on the assumed allowable bearing pressure.

Step 3: Soil Bearing Check

Average factored bearing pressure:

$$\begin{aligned} q_u &= \frac{P_u}{A_f} \\ q &= \frac{520}{13.5 \times 13.5} \\ q &= 2.85 \text{ ksf} \end{aligned}$$

$$q_{\text{allow}} = 3.0 \text{ ksf}$$

$$2.85 \text{ ksf} < 3.0 \text{ ksf} \dots \text{Bearing Pressure Pass}$$

Step 4: Footing Thickness

Adopt:

$$h = 30 \text{ in.}$$

Given:

$$B = L = 13.5 \text{ ft}$$

$$c = 39 \text{ in.}$$

$$f'_c = 4.0 \text{ ksi}$$

Bottom reinforcement:

$$\#8$$

Concrete cover:

$$3 \text{ in.}$$

For #8 bars:

$$d_b = 1.0 \text{ in.}$$

Therefore:

$$d = h - \text{cover} - \frac{d_b}{2}$$

$$d = 30 - 3 - \frac{1.0}{2}$$

$$d = 26.5 \text{ in.}$$

Step 5: Two-Way Shear / Punching Shear

For a square pedestal, the critical punching-shear perimeter is located approximately $\frac{d}{2}$ from the pedestal face.

$$d = 26.5 \text{ in.}$$

$$b_o = 4(c + d)$$

$$b_o = 4(39 + 26.5)$$

or:

$$b_o = 262 \text{ in}$$

$$b_o = \frac{262}{12} = 21.83 \text{ ft}$$

$$V_u = P_u - q_u(c + d)^2$$

$$P_u = 520 \text{ kip}$$

$$q_u = 2.853 \text{ ksf}$$

Area inside the critical perimeter:

$$A_o = (c + d)^2$$

$$A_o = (39 + 26.5)^2$$

$$A_o = 4290.25 \text{ in}^2 = 29.79 \text{ ft}^2$$

Soil reaction inside:

$$P_{soil} = q_u A_o$$

$$P_{soil} = 2.853(29.79)$$

$$P_{soil} = 84.99 \text{ kip}$$

Therefore:

$$V_u = 520 - 84.99$$

$$V_u = 435.0 \text{ kip}$$

For square interior pedestal:

Therefore:

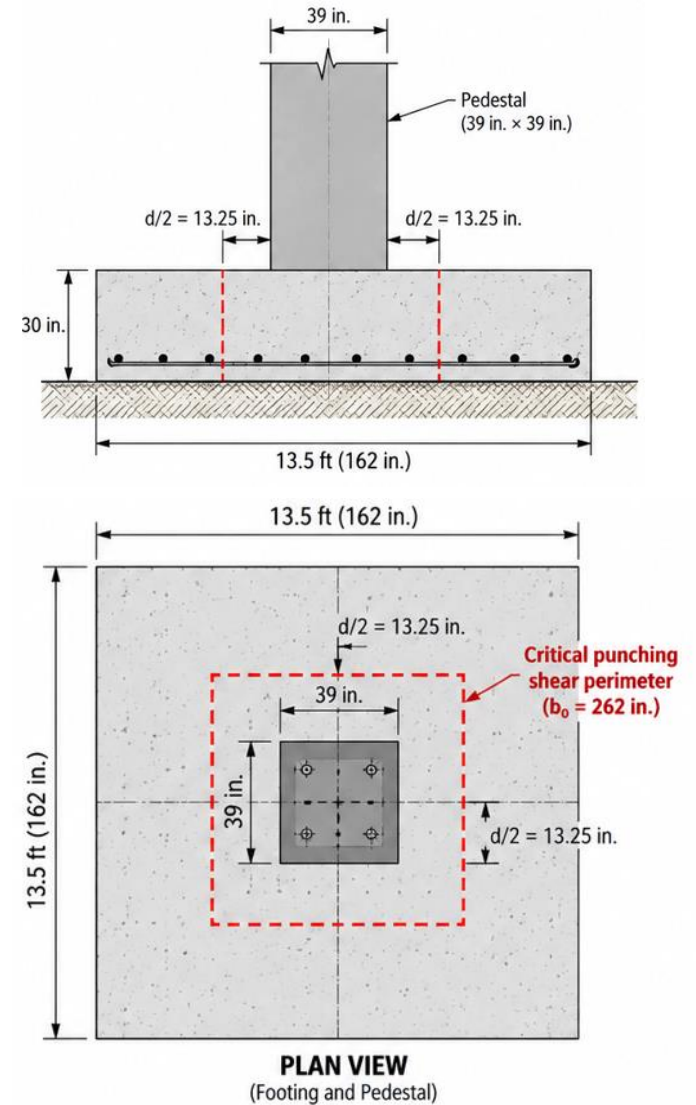
$$\beta = \frac{39}{39} = 1.0$$

So $\beta = 1$ means the loaded area is **square**.

$$\beta = 1$$

location of pedestal

α_s depends on where the loaded area is located (ACI 318-19 §22.6.5.3):



Pedestal location α_s

Interior **40**

Edge **30**

Corner **20**

Your footing is being designed as an interior/square pedestal condition, so:

$$\alpha_s = 40$$

Equation 1:

$$\phi V_c = 0.75 \left(4\lambda_s \lambda \sqrt{f'_c} \right) b_o d$$

However, for **isolated footings**, the size-effect factor may be taken as:

$$\lambda = 1.0$$

The size-effect factor may be neglected for shallow foundations based ACI-318-19.

Thus:

$$\begin{aligned} \phi V_c &= 0.75(4)(1)(1)\sqrt{4000}(262)(26.5) \\ \phi V_{c1} &= 1317.3 \text{ kip} \end{aligned}$$

The other two expressions give approximately:

Equation 2:

$$\begin{aligned} \phi V_{c2} &= 0.75 \left(2 + \frac{4}{1} \right) (1)(1)\sqrt{4000}(262)(26.5) \\ \phi V_{c2} &= 1976.0 \text{ kip} \end{aligned}$$

Equation 3

$$\begin{aligned} \phi V_{c3} &= 0.75 \left(2 + \frac{40(26.5)}{262} \right) (1)(1)\sqrt{4000}(262)(26.5) \\ \phi V_{c3} &= 1991.1 \text{ kip} \end{aligned}$$

Therefore:

$$\phi V_c = 1317.3 \text{ kip}$$

Check:

$$435.0 < 1317.3 \dots \text{PUNCHING SHEAR — PASS}$$

Utilization:

$$\frac{435.0}{1317.3} = 0.330$$
$$\text{Utilization} = 33.0\%$$

Step 7: One-Way Shear

$$B = L = 13.5 \text{ ft}$$

$$c = 39 \text{ in} = 3.25 \text{ ft}$$

$$d = 26.5 \text{ in} = 2.208 \text{ ft}$$

The footing projection is:

$$m = \frac{L - c}{2}$$
$$m = \frac{13.5 - 3.25}{2}$$
$$m = 5.125 \text{ ft}$$

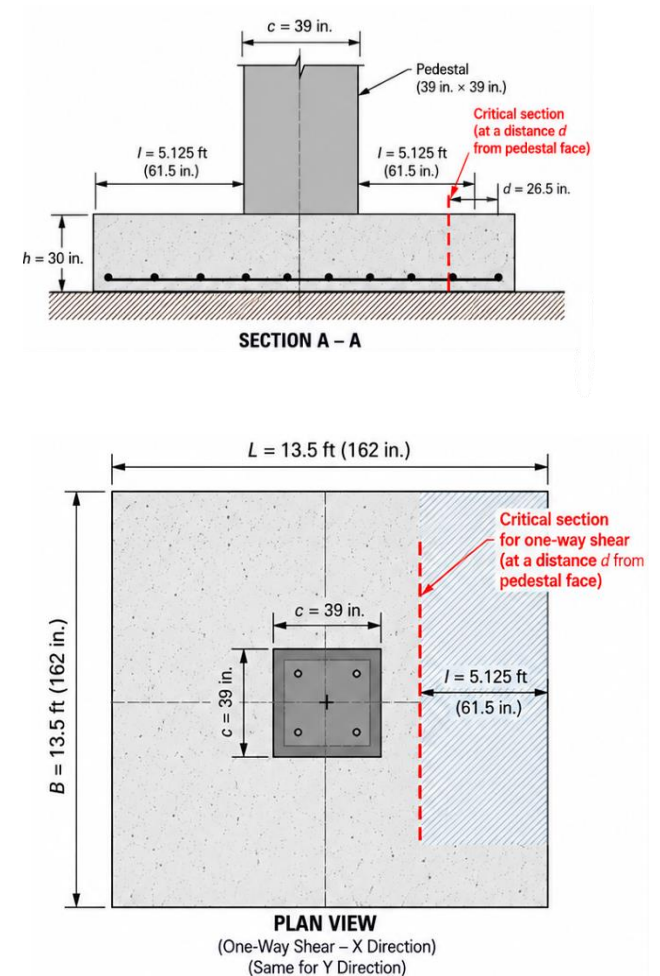
But the critical shear section is d from the pedestal face, not at the end of the projection.

Therefore:

$$V_u = q_u B \left(\frac{L}{2} - \frac{c}{2} - d \right)$$
$$V_u = 2.853(13.5)(6.75 - 1.625 - 2.208)$$
$$V_u = 112.35 \text{ kip}$$

One-way shear strength

$$\phi V_c = 0.75 \left[8\lambda(\rho_w)^{1/3} \sqrt{f'_c} \right] b_w d$$



with:

$$\rho_w = \frac{A_s}{b_w d}$$

and:

$$\rho_w \geq 0.0018$$

Using adopted bottom:

$$\#8@8"$$

$$A_s = 0.79 \left(\frac{12}{8} \right)$$

$$A_s = 1.185 \text{ in}^2/\text{ft}$$

Thus:

$$\rho_w = \frac{1.185}{12(26.5)}$$

$$\rho_w = 0.00373$$

which is greater than:

$$0.0018$$

Therefore:

$$V_c = 8(1)(0.00373)^{1/3} \sqrt{4000}(162)(26.5)$$

$$V_c \approx 336.8 \text{ kip}$$

and:

$$\phi V_c = 0.75(336.8)$$

$$\phi V_c = 252.6 \text{ kip}$$

Check:

$$112.35 < 252.6 \dots \dots \text{ONE-WAY SHEAR — PASS}$$

Utilization:

$$\frac{112.35}{252.6} = 0.445$$

$$\mathbf{44.5\%}$$

Step 8: Flexure and Reinforcement

The cantilever projection is:

$$m = \frac{L - c}{2} = 5.125 \text{ ft}$$

Therefore:

$$M_u = q_u B \left(\frac{L}{2} - \frac{c}{2} \right)^2 / 2$$

or per 1-ft strip:

$$M_u = \frac{q_u m^2}{2}$$

$$M_u = \frac{2.853(5.125)^2}{2}$$

$$M_u = 37.43 \text{ kip-ft/ft}$$

$$A_{s,min} = 0.0018A_g$$

For a 12-in strip:

$$A_g = 12(30) = 360 \text{ in}^2$$

$$A_{s,min} = 0.0018(360)$$

$$A_{s,min} = 0.648 \text{ in}^2/\text{ft}$$

Provided with #8 @ 8":

$$A_s = 0.79(12/8)$$

$$A_s = 1.185 \text{ in}^2/\text{ft}$$

Therefore:

1.185 > 0.648 Minimum reinforcement — PASS

$$s_{max} = \min(60, 18) = 18 \text{ in}$$

$$8 < 18 \dots \dots \text{PASS.}$$

Step 9: Pedestal/Footing Bearing

$$N_1 = \phi(0.85f'_cA_1)$$

with:

$$\phi = 0.65$$

For your 39×39 pedestal:

$$A_1 = 39(39) = 1521 \text{ in}^2$$

Therefore:

$$N_1 = 0.65(0.85)(4)(1521)$$

$$N_1 = 3361.4 \text{ kip}$$

Compare:

$$P_u = 520 \text{ kip}$$

$$520 < 3361.4 \dots \dots \text{PEDESTAL BEARING — PASS}$$

Step 10: Dowel / Pedestal-to-Footing Reinforcement

The lecture requires:

$$A_{s,dowel} \geq 0.005A_1$$

and at least four dowels.

For your pedestal:

$$A_{s,min} = 0.005(1521)$$

$$A_{s,min} = 7.605 \text{ in}^2$$

Your adopted 18-#6:

$$A_s = 18(0.44)$$

$$A_s = 7.92 \text{ in}^2$$

Therefore:

$$7.92 > 7.605 \dots \dots \text{DOWEL/INTERFACE STEEL — PASS}$$

Step 11: Development Length

$$l_{dc} = \frac{f_y \psi_r}{50 \lambda \sqrt{f'_c}} d_b$$

with the minimum requirement:

$$0.0003 f_y d_b \geq 8 \text{ in.}$$

For #6:

$$d_b = 0.75 \text{ in.}$$

$$l_{dc} = \frac{60,000(1.0)}{50(1)(\sqrt{4000})} (0.75)$$

$$l_{dc} \approx 14.23 \text{ in.}$$

Minimum:

$$0.0003(60,000)(0.75) = 13.5 \text{ in.}$$

Therefore:

$$l_{dc} = 14.23 \text{ in}$$

For detailing, we can adopt:

18 in. minimum developed length

